

Lab 15: Regularization and Ridge Regression

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15.1 NBA Lineup Data and Player Effects

Use `15_nba-lineups.rds` to estimate player impact from 2023–24 NBA possession data. Each row is one possession, with outcome `pts_pos`, offensive lineup `lineup_team`, and defensive lineup `lineup_opp`.

The starter code handles data preparation and chronological training, validation, and test splits. Your work should focus on fitting and comparing the plus-minus models.

The first plus-minus design gives player j one total-impact coefficient β_j . For possession i , define

$$x_{ij}^{\text{total}} = \begin{cases} 1, & \text{player } j \text{ is on offense on possession } i, \\ -1, & \text{player } j \text{ is on defense on possession } i, \\ 0, & \text{otherwise.} \end{cases}$$

The model is

$$y_i = \alpha_0 + \sum_{j=1}^m x_{ij}^{\text{total}} \beta_j + \epsilon_i.$$

Positive coefficients mean that, after accounting for the other players on the floor, the player's team tends to score more on offense and allow fewer points on defense.

The second version gives each player separate offensive and defensive coefficients:

$$\beta_j^{\text{off}} \quad \text{and} \quad \beta_j^{\text{def}}.$$

Use the lecture sign convention:

$$y_i = \alpha_0 + \sum_{j \in OP(i)} \beta_j^{\text{off}} - \sum_{j \in DP(i)} \beta_j^{\text{def}} + \epsilon_i.$$

Larger offensive coefficients mean stronger scoring impact. Larger defensive coefficients mean stronger point prevention because they are subtracted from the offense's points.

15.2 Regularization and Model Selection

Fitting the plus-minus model by ordinary least squares gives adjusted plus-minus (APM). Basketball lineup data are highly collinear, so APM can give extreme estimates to players who share most possessions with the same teammates.

Regularized adjusted plus-minus (RAPM) uses ridge regression:

$$\hat{\beta}^{\text{ridge}} = \arg \min_{\beta} \left\{ \sum_i (y_i - x_i^T \beta)^2 + \lambda \sum_j \beta_j^2 \right\}.$$

The tuning parameter λ controls how strongly player effects are pulled toward zero.

Choose λ using the prepared held-out split:

1. Fit candidate ridge models on the training games over a grid of λ values.
2. Choose λ using validation RMSE.
3. Evaluate on the test games only after the modeling procedure is finalized.

Do not use the test set while choosing λ , selecting players for plots, or deciding how to change the model.

15.2.1 Task 1: Fit Combined Plus-Minus Models

Using the prepared design matrix `X_combined`, fit:

1. a mean-only baseline;
2. raw plus-minus for each player;
3. unregularized APM;
4. ridge RAPM over a grid of candidate λ values.

Report the validation RMSE for each choice of λ , and choose the minimizing λ .

15.2.2 Task 2: Visualize Shrinkage

Choose at least ten players to show. Plot raw plus-minus, APM, and RAPM estimates from the combined model. Report estimates in points per 100 possessions and include a zero line. Sort the plot by RAPM.

Identify which players' estimates move most under ridge regularization. Why might this be? Support your explanation with data.

15.2.3 Task 3: Fit an Offense-Defense RAPM Model

Using the prepared split design matrix `X_split`, fit ridge models over the same λ grid and choose λ using validation RMSE. Make a plot for your selected players showing offensive and defensive estimates separately in points per 100 possessions. Identify at least one player whose total-impact estimate hides a meaningful offense-defense difference.

15.2.4 Task 4: Final Test Evaluation

After all choices above are fixed, evaluate the final models on the test games. Report test RMSE for the mean-only baseline, raw plus-minus, combined APM, combined RAPM, and split RAPM.

Which model predicts held-out possessions best? Explain what the split model reveals that the combined model hides, and whether the extra flexibility appears useful for prediction.